

## Introduction

A Kaplan-Meier survival curve is the most recognisable figure in clinical biostatistics. It estimates the probability of surviving (or remaining event-free) past each time point and displays the result as a step function dropping at every observed event. A publication-grade KM plot is rarely just the curve, however — it carries a risk table beneath the panel showing how many subjects remain at each landmark time, a log-rank  $p$ -value annotating the between-group comparison, and 95 % confidence bands shaded behind the curve. The `survminer` package combines all of these into a single `ggsurvplot()` call that works directly on `survfit` objects from the `survival` package.

## Prerequisites

A working understanding of survival analysis, the Kaplan-Meier estimator, censoring, and the log-rank test. Familiarity with `ggplot2` helps when customising the resulting object.

## Theory

The KM estimator is

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right),$$

where  $d_i$  is the number of events and  $n_i$  the number at risk just before time  $t_i$ . The plot displays  $\hat{S}(t)$  on the y-axis, time on the x-axis, with steps at each event and tick-marks (or shorter steps) at censoring times. Multiple groups are overlaid in different colours, and the risk-table below the panel shows  $n_i$  at chosen landmark times for each group. The log-rank  $p$ -value summarises the between-group difference under proportional hazards.

## Assumptions

Right-censored survival data, non-informative censoring (the censoring mechanism does not depend on prognosis), and — for the log-rank annotation — approximately proportional hazards across groups. Severely non-proportional hazards make the log-rank test underpowered and a restricted-mean comparison may be more appropriate.

## R Implementation

```
library(survival); library(survminer)

fit <- survfit(Surv(time, status) ~ sex, data = lung)

ggsurvplot(fit,
  data = lung,
  conf.int = TRUE,
  pval = TRUE,
  risk.table = TRUE,
  legend.labs = c("Male", "Female"),
  palette = c("#2A9D8F", "#F4A261"),
  xlab = "Days",
  ylab = "Survival probability",
  risk.table.y.text.col = TRUE
)
```

```
# Cumulative event curve (1 - S)
ggsurvplot(fit, data = lung, fun = "event")
```

## Output & Results

A two-panel figure: the upper panel shows two survival curves (male and female) with shaded 95 % confidence bands and a log-rank  $p$ -value in the corner; the lower panel is a risk table tabulating the number at risk in each group at six evenly spaced time points. The cumulative-event variant (`fun = "event"`) inverts the y-axis to show  $1 - \hat{S}(t)$ , useful when the event rate is the more natural reporting scale.

## Interpretation

A reporting sentence (figure caption): “Kaplan–Meier overall-survival curves by sex (`survival::survfit`); shaded bands are 95 % point-wise confidence intervals, the log-rank  $p$ -value compares the two groups, and the risk table reports the number of subjects at risk at each labelled time.” Always state the time-zero anchor (date of randomisation, surgery, diagnosis) in the methods so the x-axis is unambiguous.

## Practical Tips

- Always include a risk table; without it, the curve at late time points is misleading because the estimate is based on a tiny remaining cohort.
- Use `fun = "event"` for cumulative incidence framing when communicating to clinical audiences who think in event rates rather than survival probabilities.
- For competing risks, switch to a cumulative incidence function (CIF) plot via `cmprsk::cuminc()` and `ggcompetingrisks()`; KM curves overestimate the event of interest when competing events are non-trivial.
- Pair the log-rank  $p$ -value with a hazard-ratio estimate (e.g. from a univariate Cox model); the test alone gives no effect size.
- For multi-arm trials, lay out subgroups in panels (`facet.by = "subgroup"`) rather than crowding many curves into one panel.
- Export at 300 dpi (or higher for line art) and at the journal’s column width so the risk-table text remains legible.

## R Packages Used

`survival` for `survfit()` and `Surv()`; `survminer` for `ggsurvplot()`, `ggsurvtable()`, and risk-table support; `ggplot2` for downstream customisation of the returned object; `cmprsk` for competing-risks extensions when KM is not appropriate.

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

## See also — labs in this chapter

- `ggplot2` grammar and multi-panel layouts